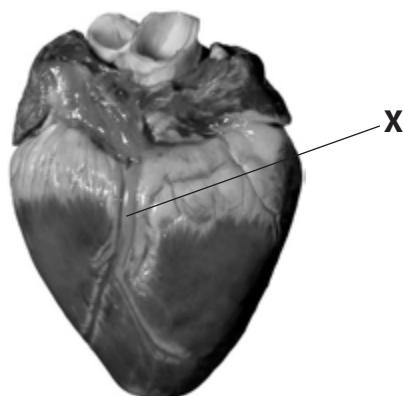


- 4 The photograph shows a mammalian heart.



The blood vessel labelled **X** is a branch from the aorta.

The aorta is one of the major blood vessels of the heart.

- (a) Explain why it is important that blood vessel **X** branches directly from the aorta.

(3)

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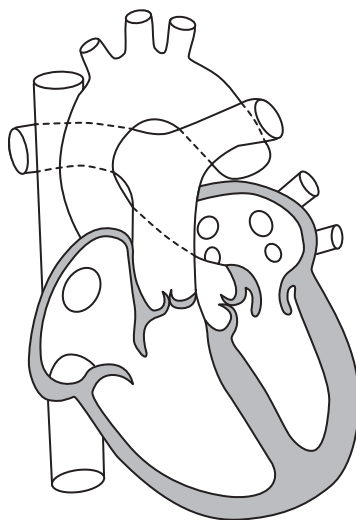
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(b) The diagram shows the internal structure of a mammalian heart.



(i) Explain which stage of the cardiac cycle is shown in this diagram.

(2)

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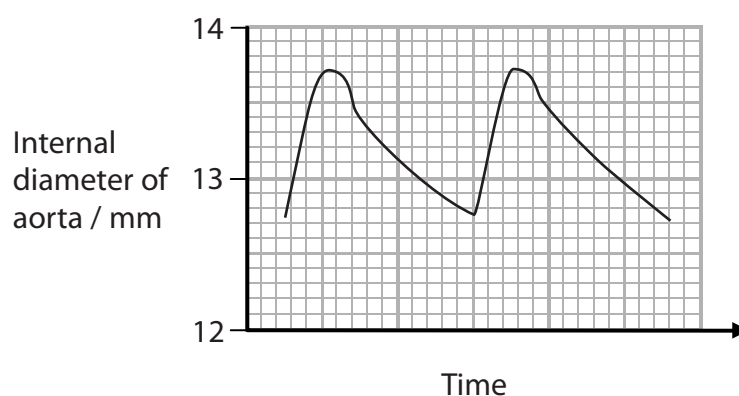
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(ii) Draw arrows on the diagram to show the flow of blood through the left side of the heart and into the aorta.

(3)



- (c) The graph shows the changes in the internal diameter of the aorta during two cardiac cycles.



Explain the changes in the internal diameter of the aorta.

(3)

(Total for Question 4 = 11 marks)